How long will it take?

5a

Speed up your calculations by using a calculator.

The table shows sample speeds for a variety of activities.

Activity	Speed (km/h)	Speed (m/s)
Boing 747 plane flying at 10 km altitude	900	
Space shuttle revolving Earth in orbit	25 000	
Truck travelling at speed limit	100	
Car driving at town speed limit	60	
Car travelling at suburban speed limit	50	
Car racing during the Bathurst 1000	150	
Cheetah running at full speed	80	
Person walking	5	
Person jogging	10	
Marathon runner racing	20	
Short distance runner racing	10	
Cyclist pedaling along	25	
Tour de France cyclist racing	40	

When you want to change a speed from kilometres per hour to metres per second, you substitute 1000 metres for the kilometre and 3600 seconds for the hour.

Let's convert the speed 2 km/h to a speed in m/s:

2 km/h = 2 . 1000m/3600s = 2 . 1000/3600 m/s = 2 . 1/3.6 m/s = 0.56 m/s

So, walking slowly and covering 2 km in 1 h is walking at a speed of 0.56 m in 1 s.

1 Follow the reverse of the process above to convert the speed of 2 m/s to km/h. (Remember that 1 m equals 1/1000 km and that 1 s is 1/3600 h.)

In summary:

- To convert km/h to m/s, divide the km/h speed by 3.6 (? km/h = ?/3.6 m/s)
- To convert m/s to km/h, multiply the m/s speed by 3.6 (? m/s = ? . 3.6 km/h)

2 Calculate the m/s values for the activities listed in the table at the top of this page, and record them in the third column.

The table below shows the sizes of five distances.

Distance	Size in km	Size in m
The distance across a sports field		100
The length of a Sydney Harbour Bridge span		500
Sydney Harbour Tunnel length	2.3	
Melbourne to Canberra distance	1000	
Canberra to Sydney distanced	300	

3 Calculate and record the missing measurements in the table above.

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How long will it take?

5b

4 State a formula for calculating **speed**, given the **distance** travelled and the **time** taken to travel that distance. Use the bold terms.

5 Rearrange this formula for speed to make time the subject.

6 Select five different speeds from the first table on the previous page. Write them down the left side of the chart below. Now select three different distances from the second table on the previous page. Write them across the top row. Calculate how long it would take to travel each distance at each speed.

		Distances		
Speeds				

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